

JEE MAIN REVISION MASTER: ALDEHYDE AND KETONE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 22 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Carbonyl Nucleophilic Addition & Ammonia Derivatives	12 PYQs	Reactions & Mechanisms
High	Named Condensation Reactions & Diagnostic Tests	11 PYQs	Named Reactions & Tests

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Carbonyl Nucleophilic Addition & Ammonia Derivatives	$>\text{C=O} + \text{H}_2\text{N-Z} \xrightarrow{\text{H}^+} >\text{C=N-Z} + \text{H}_2\text{O}$
Named Condensation Reactions & Diagnostic Tests	$2\text{CH}_3\text{CHO} \xrightarrow{\text{dil. NaOH}} \text{CH}_3\text{CH(OH)CH}_2\text{CHO} \xrightarrow{\Delta} \text{CH}_3\text{CH=CHCHO}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Carbonyl Nucleophilic Addition & Ammonia Derivatives (High Yield — 12 PYQs)

Core Principle: Nucleophilic addition to $>\text{C=O}$: addition of HCN gives cyanohydrins, addition of NaHSO_3 gives crystalline bisulphite adducts. Reaction with ammonia derivatives $\text{H}_2\text{N-Z}$ eliminates water to give crystalline derivatives (oximes with NH_2OH , hydrazones with NH_2NH_2 , 2,4-DNP derivatives).

■ **Caution / JEE Pitfall:** In semicarbazide $\text{H}_2\text{N-CO-NH-NH}_2$, only the hydrazine NH_2 reacts; the amide NH_2 is deactivated by resonance with $>\text{C=O}$.

■ **Speed Trick:** Reactivity order toward nucleophilic addition: $\text{HCHO} > \text{RCHO} > \text{RCOR}$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "The major product 'A' in the given reaction is Benzaldehyde $\rightarrow$ $\leq$ $\rightarrow$ 'A' (Major product)..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Match the reagents in List-I with products obtained from their carbonyl compounds in List-II. List-I List-II (a) NH_2OH (i...) [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Arrange the following compounds in their decreasing order of reactivity towards Nucleophilic addition reaction. $\text{CH}_3\text{COCH}_3, \text{CH}_3\text{COC}_2\text{H}_5$..." [Ans: N/A]

3.2 Named Condensation Reactions & Diagnostic Tests (High Yield — 11 PYQs)

Core Principle: Aldol condensation requires α -hydrogen. Cannizzaro occurs for aldehydes lacking α -H (disproportionation into alcohol + carboxylate). Iodoform test: positive for compounds containing $\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CH}(\text{OH})-$ group. Tollens' & Fehling's tests oxidize aldehydes, not ketones.

■ **Caution / JEE Pitfall:** Acetophenone ($\text{C}_6\text{H}_5\text{COCH}_3$) gives positive iodoform test, while benzophenone does not.

■ **Speed Trick:** Clemmensen uses acidic conditions (Zn-Hg / conc. HCl); Wolff-Kishner uses basic conditions (NH_2NH_2 / KOH / glycol).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The compound that does not answer iodoform test is..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "Compounds P and R in the following reaction are $\text{CH}_3\text{CHO} \xrightarrow[\text{conc. H}_2\text{SO}_4]{\text{H}_2\text{O}^+} \text{CH}_3\text{MgBr} \xrightarrow{\text{Heat}}$ " [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "The test to differentiate between pentan-2-one and pentan-3-one is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The compound that does not answer iodoform test is

- (A) Ethanal
- (B) Acetone
- (C) Ethanoic acid
- (D) Acetophenone

Correct Answer: None

Solution Analysis: The compound that does not answer the iodoform test is: Option C: Ethanoic acid. Iodoform test is given by compounds containing the group $-\text{COCH}_3$. This is called the methyl ketone group. Some alcohols which can be oxidised to methyl ketones also give this test. Now check the options: Ethanal has the structure: CH_3CHO . It gives iodoform test. Acetone has the structure: CH_3COCH_3 . It contains the methyl ketone group, so it gives iodoform test. Acetophenone has the structure: $\text{C}_6\text{H}_5\text{COCH}_3$. It also contains the methyl ketone group, so it gives iodoform test. Ethanoic acid

Benchmark Problem 2: KCET 2026 (Q#2)

The major product 'A' in the given reaction is Benzaldehyde $\rightarrow \text{C}_6\text{H}_5\text{CHO}$ (Major product)

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: This reaction is a crossed aldol condensation. Since it occurs between benzaldehyde and acetophenone in the presence of a base, it is also called Claisen-Schmidt condensation. Benzaldehyde ($\text{C}_6\text{H}_5\text{CHO}$) does not have any α -hydrogen. So, it cannot form an enolate ion. Acetophenone ($\text{C}_6\text{H}_5\text{COCH}_3$) has α -hydrogens, so the base removes one α -hydrogen from it. This gives the enolate ion of acetophenone:

